

Lunar Exploration by Privately-Funded Teams: A New Beginning



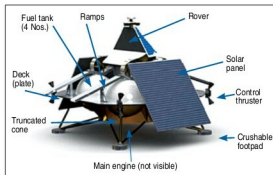
Subodh P. Kachhela is a retired scientist/engineer, Space Applications Centre, ISRO, Ahmedabad

Space missions across the world have mainly been funded by respective government agencies. The first phase of private space operation in the USA was the launch of commercial communication satellites, thanks to the US Communication Satellite Act of 1962. Missions for space exploration, however, continued to be government-funded. Historically, SpaceShipOne was the spaceplane that completed the first manned private spaceflight in 2004, which won the US\$ 10 million Ansari X prize but was immediately retired from active service.

Augustine Committee was set up to review human spaceflight plans of the USA. In its review, in 2009, it took into account several objectives like support for International Space Station (ISS), development of missions beyond low-Earth orbit (including Mars, moon and near-Earth objects) and use of commercial space industry.

After this, SpaceX (Space Exploration Technologies Corp.) became the first privately-owned company that built unmanned, reusable Dragon Space capsule. The capsule was ferried to space atop the company's Falcon9 rocket launched from Cape Canaveral on May 22, 2012, to dock with ISS. The capsule returned back to Earth after nine days. It delivered supplies to astronauts aboard.

Lander-Rover combo for Red Rover (left), and Red Rover (right)



Moon missions by private teams

Specifically for moon exploration, a contest called Google Lunar X Prize (GLXP) has been designed. It is believed that this will boost the role of private companies in other space missions and inspire innovation in spaceflight technologies.

GLXP is igniting a new era of lunar exploration by offering the largest international incentive prize of all time. A total of US\$ 30 million in prizes are available to the privately-funded teams. The competition requisites are as follows:

Landing. Safely land a robot (rover) on lunar surface

Moving. Have that rover travel 500m over the lunar surface

Imaging. Send videos, images and data back to Earth

The teams must be at least 90 per cent privately-funded, though commercially reasonable sales to government customers are allowed without limit.

Approximately 33 teams across the world have been preparing to participate in the competition by fund raising, mission planning and building robots in a new race to the moon. As per the latest information, the deadline for the competition has been extended till December 31, 2016.

Astrobotic and their robots

Prof. Whittaker established Astrobotic Technology at Pittsburgh, the USA. He has a rich experience in the field of robotics. His robots explored damaged and dangerously radioactive areas of nuclear power plants, hunted meteorites in the ice fields of Antarctica